

MAT7058 Combinatorics II, Exercises 3

Due Date: 01 April 2026, Wednesday, beginning of class.
Only handwritten paper submissions in English are accepted.

- Each exercise is worth 4 points.
 - In general, you have to prove everything and only use things which were introduced in class. Ask if you are unsure.
- 1 (a) The graph $G = (S \cup T, E)$ is bipartite and k -regular with $k \geq 1$. Show that $|S| = |T|$ and that G contains a matching M with $|M| = |S| = |T|$.
(b) A *1-factor* of a graph is a matching which contains all vertices in one of its edges. Show that any k -regular, bipartite graph possesses $k!$ 1-factors ($k \geq 1$).
 - 2 An *edge-cover* of a graph $G = (V, E)$ is a set of edges C such that every vertex is in one edge of C . The size of a smallest edge cover of G is denoted by $\rho(G)$. Let G be a connected graph. Let us denote the matching number of G by $\nu(G)$.
 - (a) Show that $\nu(G) + \rho(G) = |V|$.
 - (b) Show that $\alpha(G) = \rho(G)$ if G is bipartite. Here $\alpha(G)$ denotes the size of the largest independent set/coclique of G , i.e., the largest set of pairwise nonadjacent vertices.
 - (c) Show that the previous equality is not true for non-bipartite graphs.
 - 3 Show that if a bipartite graph $G = (S \cup T, E)$ does not have a matching of size $|S|$, then its adjacency matrix has an eigenvalue 0.
 - 4 Let $\mathcal{F} = \{F_1, F_2, \dots, F_m\}$ be a family of finite sets. Show that if

$$\sum_{1 \leq i < j \leq m} \frac{|F_i \cap F_j|}{|F_i| \cdot |F_j|} < 1,$$

then $\mathcal{F}$ has a transversal.

- 5 Let λ be a nonnegative integer and μ a positive integer with $\lambda \leq \mu$.
 - (a) Show that

$$ex(v, \{K_{1,1,\lambda+1}, K_{2,\mu+1}\}) \leq \frac{v}{4}(\lambda - \mu + 1 + \sqrt{(\lambda - \mu + 1)^2 + 4\mu(v-1)}).$$

(Hint: The case $(\lambda, \mu) = (0, 1)$ should look familiar.)

(b) Characterize equality in the bound.

- 6** Let G be a graph with n points and m edges that does not contain a $K_{r,r}$. Prove that

$$m < C \cdot n^{2-\frac{1}{r}},$$

where C only depends on r (not n or m). (Hint: Count $K_{1,r}$'s in two ways.)